

# Bitcoin

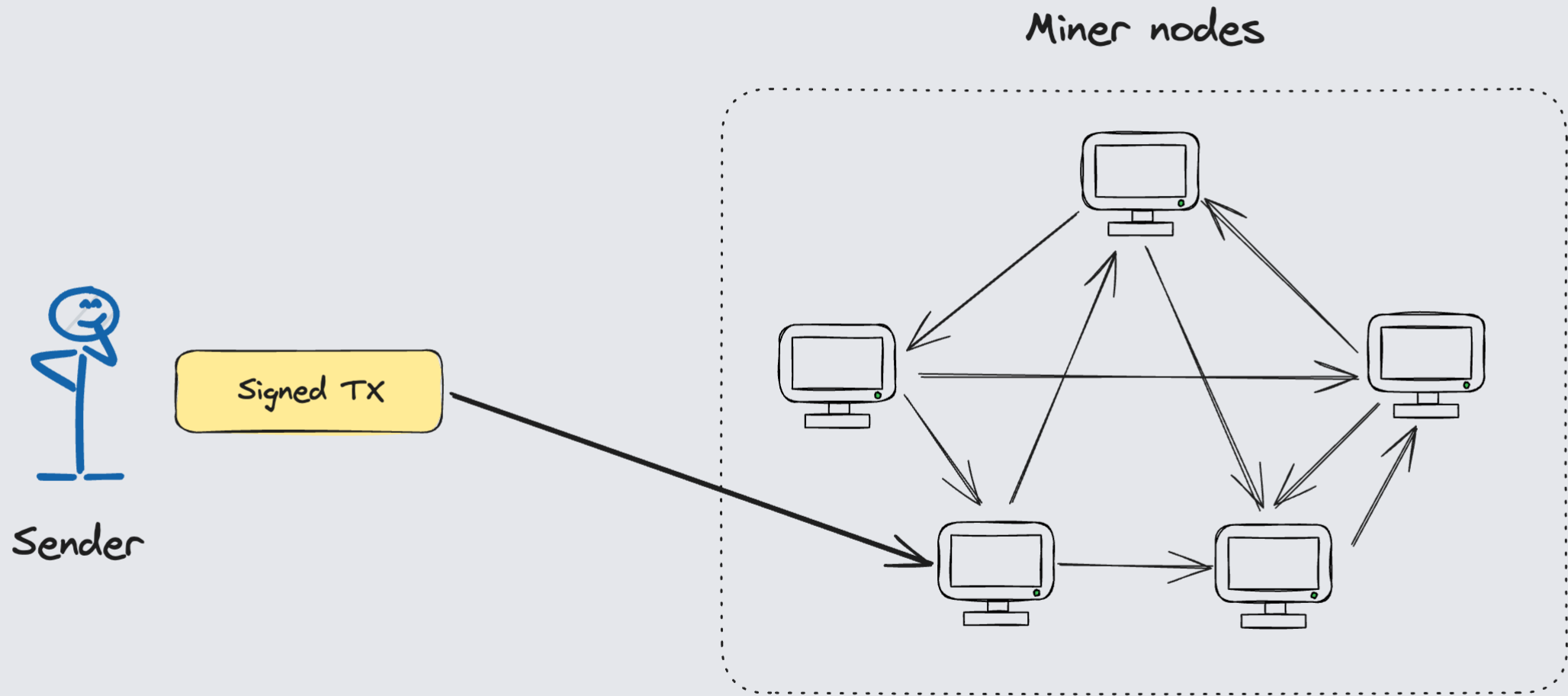
Prepared by Kirill Sizov



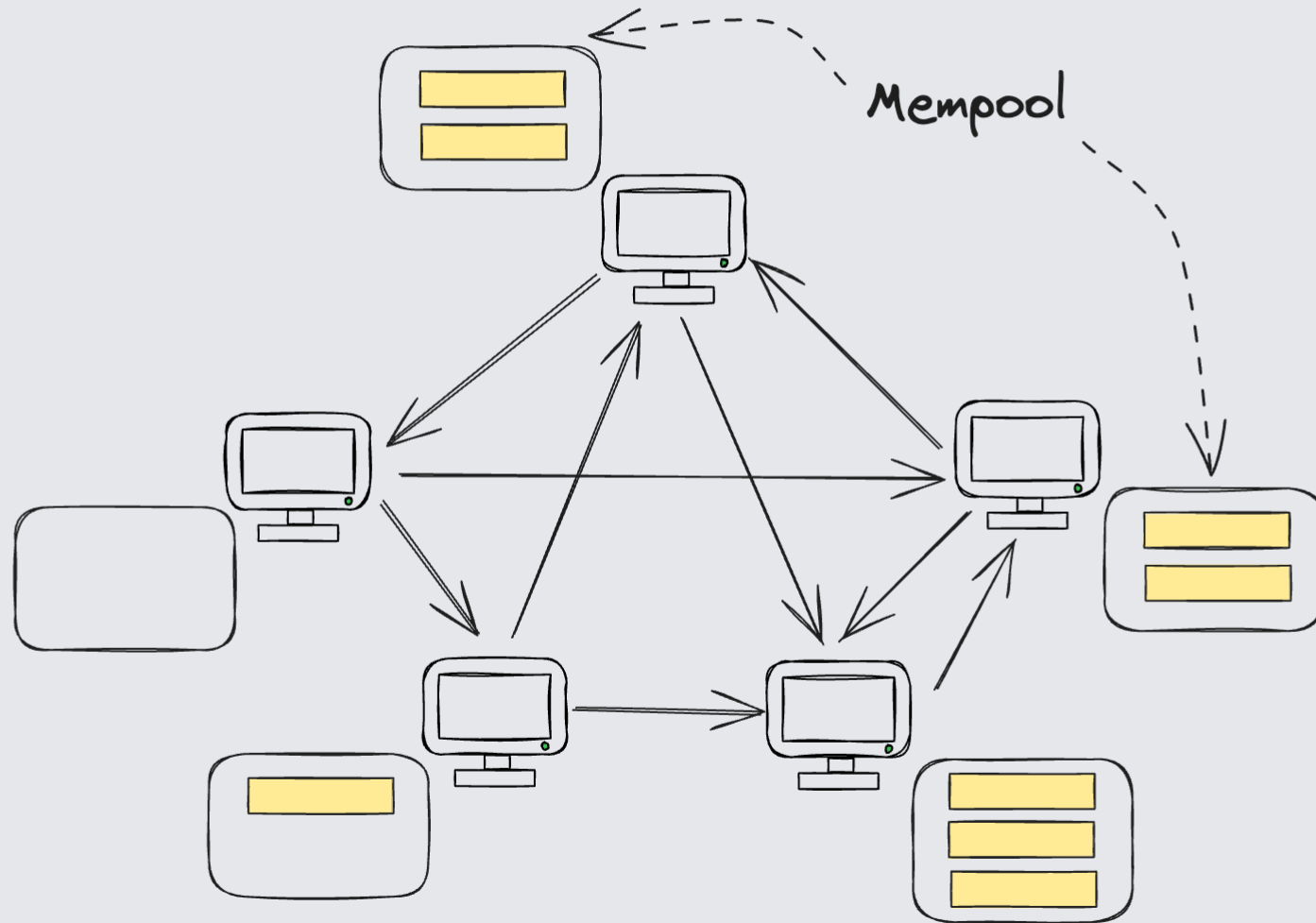
# Agenda

- Transaction flow
- Bitcoin consensus
- Block structure
- Visual demo
- Transaction structure
- Bitcoin script

# Sending a TX

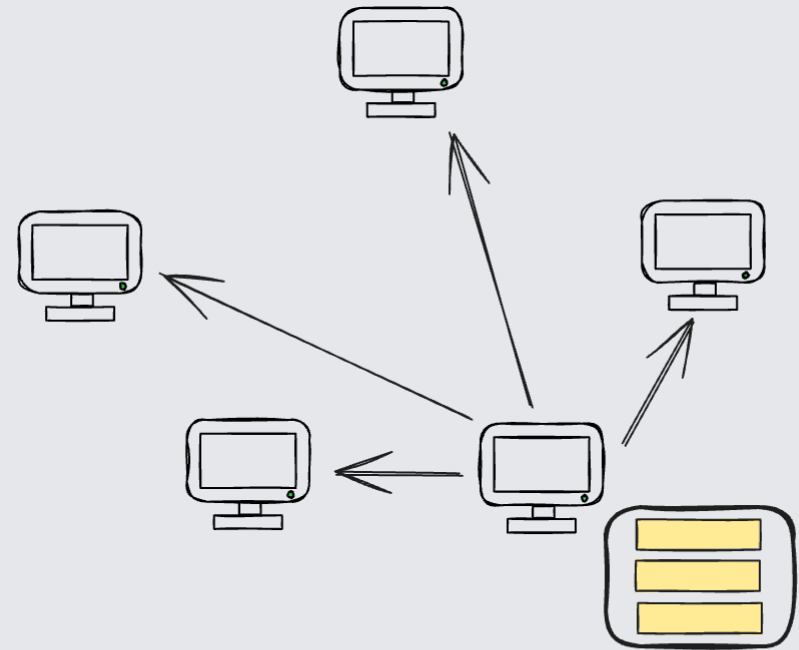


# TX propagation



# Block creation

- Each miner select TXs and build their own block.
- PoW consensus mechanism select a winner node.
- Selected node propagates their block to other nodes.
- Other nodes validate this block.



# The rule to win

- A block wins if its header hash  $H$  is  $\leq T$ , where  $T$  is the target encoded in `nBits`.
- $H = \text{SHA256}(\text{SHA256}(\text{header}))$ , with  $H \in [0, 2^{256} - 1]$
- Miners explore new hashes by varying:
  - Nonce (32-bit),
  - Coinbase (changes Merkle root),
  - Timestamp
- Accept block  $\iff H \leq T$

# Probability per hash

- Model each hash as uniform on  $[0, 2^{256} - 1]$
- Success probability for one trial:

$$p = \Pr[H \leq T] = \frac{T + 1}{2^{256}} \approx \frac{T}{2^{256}}.$$

- Expected trials for one success:

$$\mathbb{E}[\text{hashes}] = \frac{1}{p} \approx \frac{2^{256}}{T}.$$

# Target, Difficulty, Work

- Define difficulty relative to “difficulty-1” target  $T_1$ :

$$D = \frac{T_1}{T} \implies T = \frac{T_1}{D}.$$

- Expected hashes to find a block at difficulty (D):

$$\mathbb{E}[\text{hashes}] \approx D \cdot 2^{32}.$$

- Chainwork contributed by a block:

$$w \approx \frac{2^{256}}{T + 1}.$$



## nBits

Let `nBits` = `[e | m]` where (e) is the first byte (exponent) and (m) are the next 3 bytes (mantissa).

$$T = m \cdot 256^{(e-3)} = m \cdot 2^{8(e-3)}.$$

# Keeping ~10-minute blocks

- Every 2016 blocks, compute actual timespan  $\Delta t$  between those blocks.
- Target timespan  $T_0 = 2016 \times 600$  s

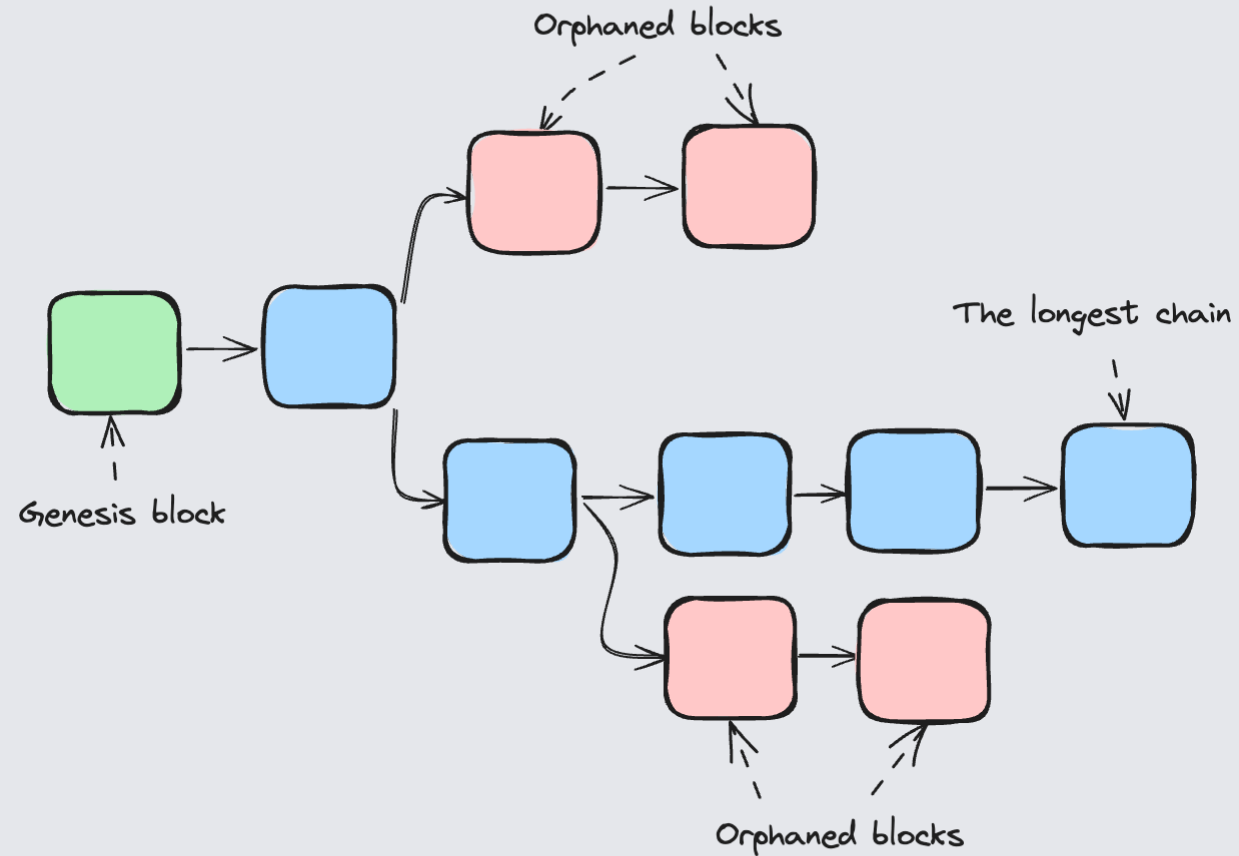
$$T_{\text{new}} = T_{\text{old}} \times \text{clamp}\left(\frac{\Delta t}{T_0}, \frac{1}{4}, 4\right).$$

- Equivalently,

$$D_{\text{new}} = \frac{D_{\text{old}}}{\text{clamp}\left(\frac{\Delta t}{T_0}, \frac{1}{4}, 4\right)}.$$

# Bitcoin consensus

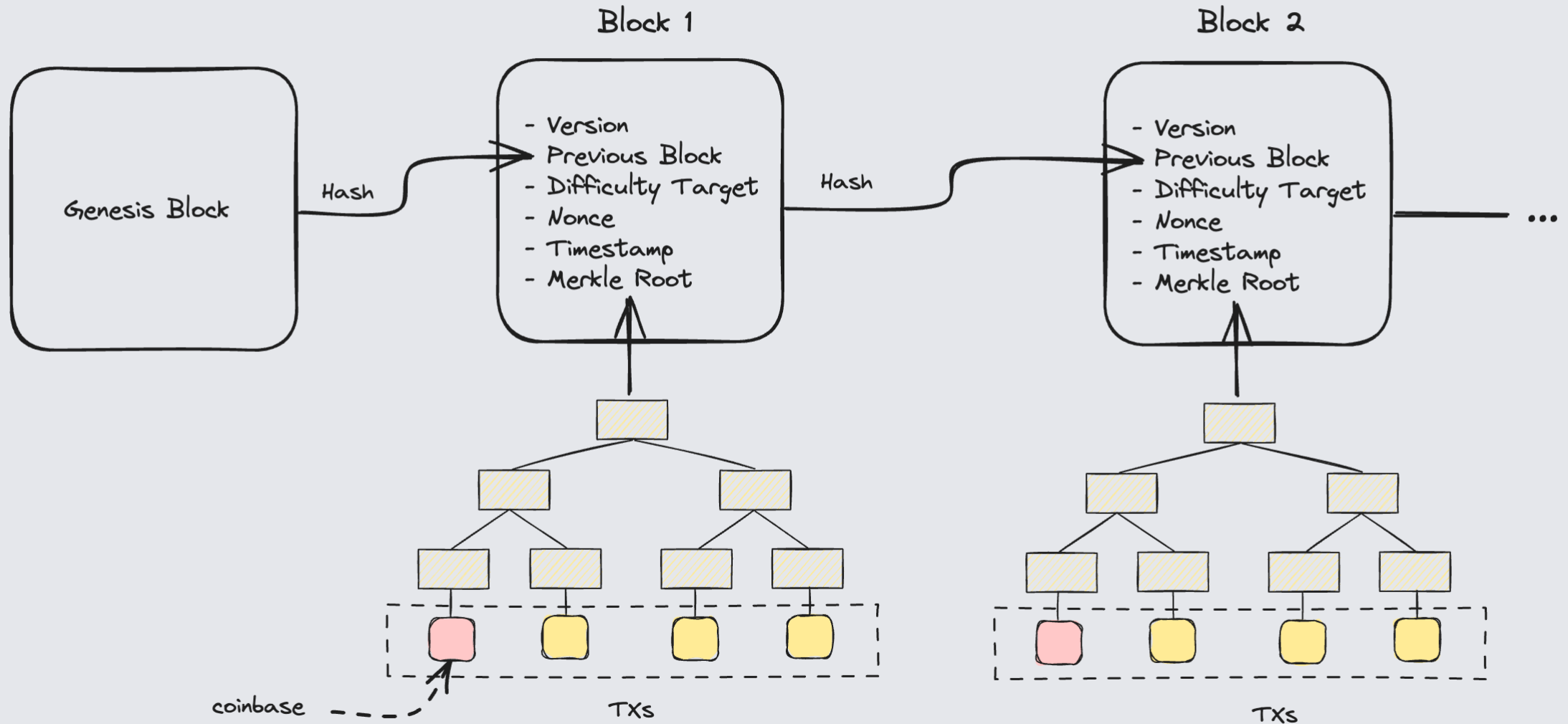
- The chain with most accumulated work (typically the longest) is considered the valid one.
- When temporary forks occur, nodes follow the chain with the most accumulated work.



Block



# Bitcoin: sequence of block headers



# Block structure

| Parameter           | Description                                     |
|---------------------|-------------------------------------------------|
| Block Size          | The size of the block in bytes.                 |
| Block Header        | A 80-byte header of the block.                  |
| Transaction Counter | The number of transactions.                     |
| Transactions        | The list of transactions included in the block. |

# Block header structure

| Parameter         | Description                                                 |
|-------------------|-------------------------------------------------------------|
| Version           | 4-byte version number.                                      |
| Previous Block    | 32-byte hash of the previous block in the blockchain.       |
| Merkle Root       | 32-byte hash based on all of the transactions in the block. |
| Timestamp         | 4-byte timestamp recording when this block was created.     |
| Difficulty Target | 4-byte number used in PoW                                   |
| Nonce             | 4-byte number used in PoW                                   |

# Explore blocks (screen from blockchain.com)

|  Latest BTC Blocks |           |         |          |          |               |         |                 |            |            |         |
|-----------------------------------------------------------------------------------------------------|-----------|---------|----------|----------|---------------|---------|-----------------|------------|------------|---------|
|                   |           |         |          |          |               |         |                 |            |            |         |
| #821147                                                                                             | #821146   | #821145 | #821144  | #821143  | #821142       | #821141 | #821140         | #821139    | #821138    | #821137 |
| #821136                                                                                             |           |         |          |          |               |         |                 |            |            |         |
| Number                                                                                              | Hash      | Miner   | Mined    | Tx Count | Nonce         | Fill    | Size            | Total Sent | Total Fees |         |
| 821146                                                                                              | 0000-b477 | Antpool | 7m 25s   | 3,921    | 1,782,447,242 | 146.78% | 1,539,112 Bytes | 12,167 BTC | 2.12BTC    |         |
| 821145                                                                                              | 0000-1455 | Antpool | 15m 35s  | 3,594    | 616,409,252   | 146.92% | 1,540,523 Bytes | 13,953 BTC | 2.82BTC    |         |
| 821144                                                                                              | 0000-8f3d | Unknown | 45m 52s  | 3,477    | 4,030,542,367 | 139.12% | 1,458,767 Bytes | 10,321 BTC | 2.17BTC    |         |
| 821143                                                                                              | 0000-4559 | Unknown | 55m 42s  | 3,104    | 597,105,437   | 141.56% | 1,484,319 Bytes | 1,998 BTC  | 2.08BTC    |         |
| 821142                                                                                              | 0000-58d6 | Antpool | 58m 51s  | 3,759    | 406,488,726   | 143.35% | 1,503,173 Bytes | 1,745 BTC  | 2.25BTC    |         |
| 821141                                                                                              | 0000-e6b1 | Unknown | 1h 2m 4s | 3,757    | 781,766,511   | 140.54% | 1,473,662 Bytes | 5,594 BTC  | 2.69BTC    |         |



# Explore blocks (screen from blockchain.com)

## Bitcoin Block 821,146

Mined on December 14, 2023 01:26:51 • [All Blocks](#)

AntPool

**Coinbase Message** • Mined by AntPool l\z#z>mmN-6S/q99#F/qOfLA y\_hgE/

A total of 12,166.78 BTC (\$521,904,659) were sent in the block with the average transaction being 3.1030 BTC (\$133,105). AntPool earned a total reward of 6.25 BTC \$268,099. The reward consisted of a base reward of 6.25 BTC \$268,099 with an additional 2.1230 BTC (\$91,067.95) reward paid as fees of the 3,921 transactions which were included in the block.

### Details

|               |                                                                                               |               |                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------|
| Hash          | 00000-8b477  | Depth         | 2                                                                                         |
| Capacity      | 146.78%                                                                                       | Size          | 1,539,112                                                                                 |
| Distance      | 17m 4s                                                                                        | Version       | 0×228d4000                                                                                |
| BTC           | 12,166.7783                                                                                   | Merkle Root   | e3-65  |
| Value         | \$521,904,659                                                                                 | Difficulty    | 67,305,906,902,031.39                                                                     |
| Value Today   | \$522,107,966                                                                                 | Nonce         | 1,782,447,242                                                                             |
| Average Value | 3.1029783864 BTC                                                                              | Bits          | 386,150,037                                                                               |
| Median Value  | 0.00665123 BTC                                                                                | Weight        | 3,993,394 WU                                                                              |
| Input Value   | 12,168.90 BTC                                                                                 | Minted        | 6.25 BTC                                                                                  |
| Output Value  | 12,175.15 BTC                                                                                 | Reward        | 8.37301099 BTC                                                                            |
| Transactions  | 3,921                                                                                         | Mined on      | 14 Dec 2023 at 13:26:51                                                                   |
| Witness Tx's  | 3,691                                                                                         | Height        | 821,146                                                                                   |
| Inputs        | 7,110                                                                                         | Confirmations | 2                                                                                         |
| Outputs       | 11,371                                                                                        | Fee Range     | 0-930 sat/vByte                                                                           |
| Fees          | 2.12301099 BTC                                                                                | Average Fee   | 0.00054145                                                                                |
| Fees Kb       | 0.0013794 BTC                                                                                 | Median Fee    | 0.00032282                                                                                |
| Fees kWU      | 0.0005316 BTC                                                                                 | Miner         | AntPool                                                                                   |

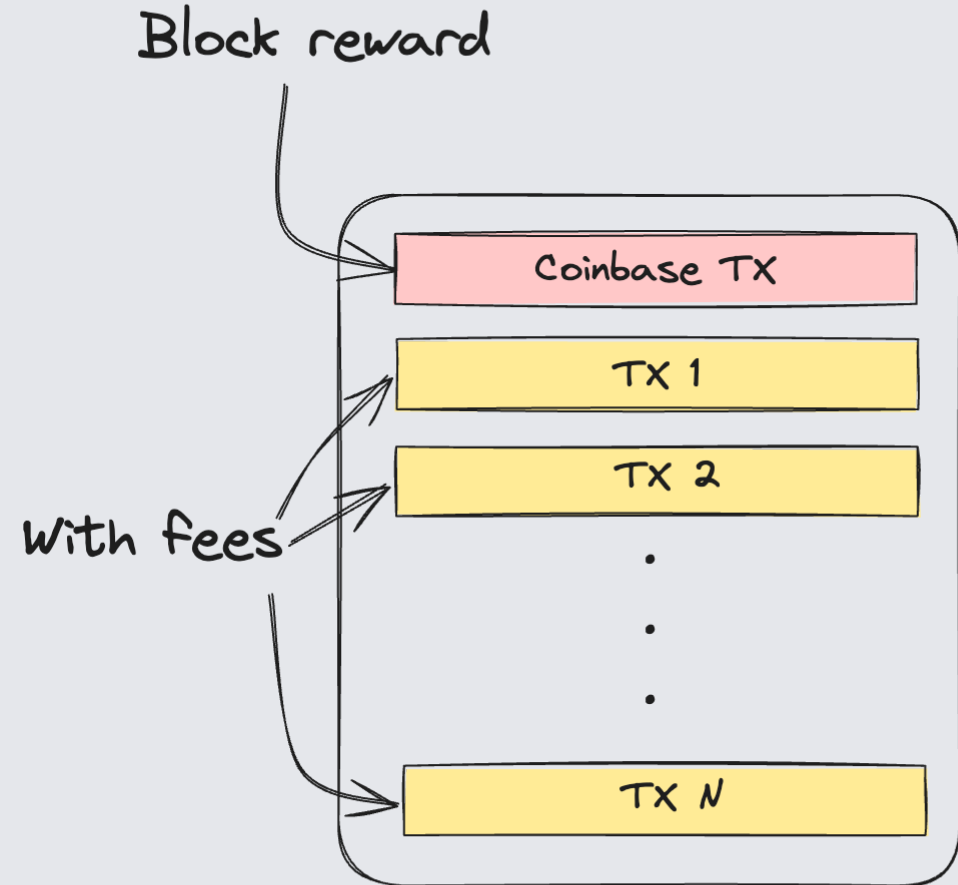
# Visual demo

Let's play with [Blockchain demo](#)

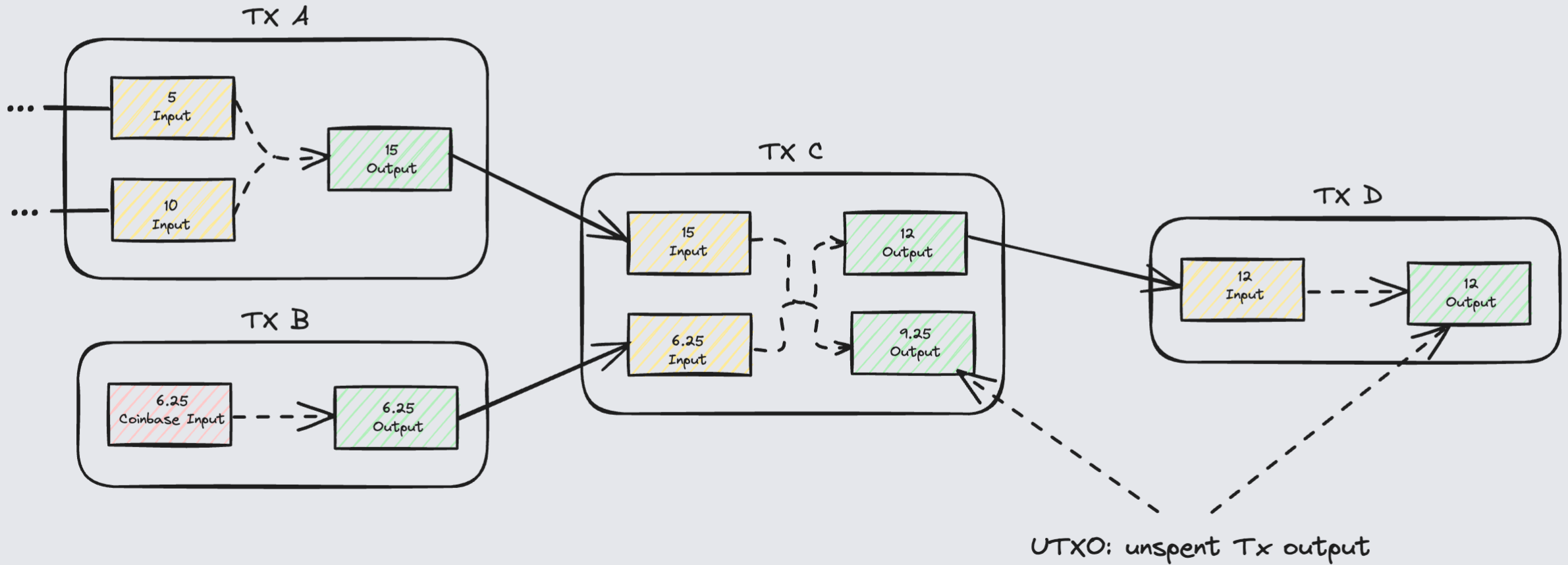
# Transactions

# Coinbase

- Miner reward consists of a block reward (coinbase) and tx fees.
- Block reward is halved after every 210,000 blocks (~4 years).
- Current reward is 3.125 BTC.



# UTXO (Unspent Transaction Output)



# TX structure

| Parameter      | Description                               |
|----------------|-------------------------------------------|
| Version        | Transaction data format version.          |
| Input Counter  | Number of transaction inputs.             |
| Inputs         | The list of transaction inputs.           |
| Output Counter | Number of transaction outputs.            |
| Outputs        | The list of transaction outputs.          |
| Locktime       | Earliest block number that can include Tx |

# TX input

| Parameter           | Description                                                         |
|---------------------|---------------------------------------------------------------------|
| Previous TX         | The hash of the transaction containing spending output              |
| Output Index        | The index of the spending output in the TX outputs                  |
| Script Length       | The length of the input script.                                     |
| Signature<br>Script | A script which provides data to the previous output's scriptPubKey. |
| Sequence            | Used for relative time locks (BIP-68/112)                           |

# TX output

| Parameter        | Description                                                           |
|------------------|-----------------------------------------------------------------------|
| Value            | The number of Satoshis to spend to this output.                       |
| Script<br>Length | The length of the output script.                                      |
| Pubkey<br>Script | A script which dictates the conditions required to spend this output. |



# TX validation

For each input in transaction, a miner node check:

- `Signature script | Pubkey script` returns true.
- `Previous TX | Output Index` is in the UTXO set.
- Previous TX is in the block.
- $\sum inputs \geq \sum outputs$

After a tx is executed, all its inputs are removed from UTXO set.

**Script language**

# Bitcoin Script

- Programming language that is used to define the conditions under which UTXO can be spent.
- Stack-based, composed of opcodes.
- Intentionally not Turing-complete, with no loops.
- <https://en.bitcoin.it/wiki/Script>

# Pay-to-Pubkey Hash (P2PKH)

One of the most common form of transaction on the Bitcoin.

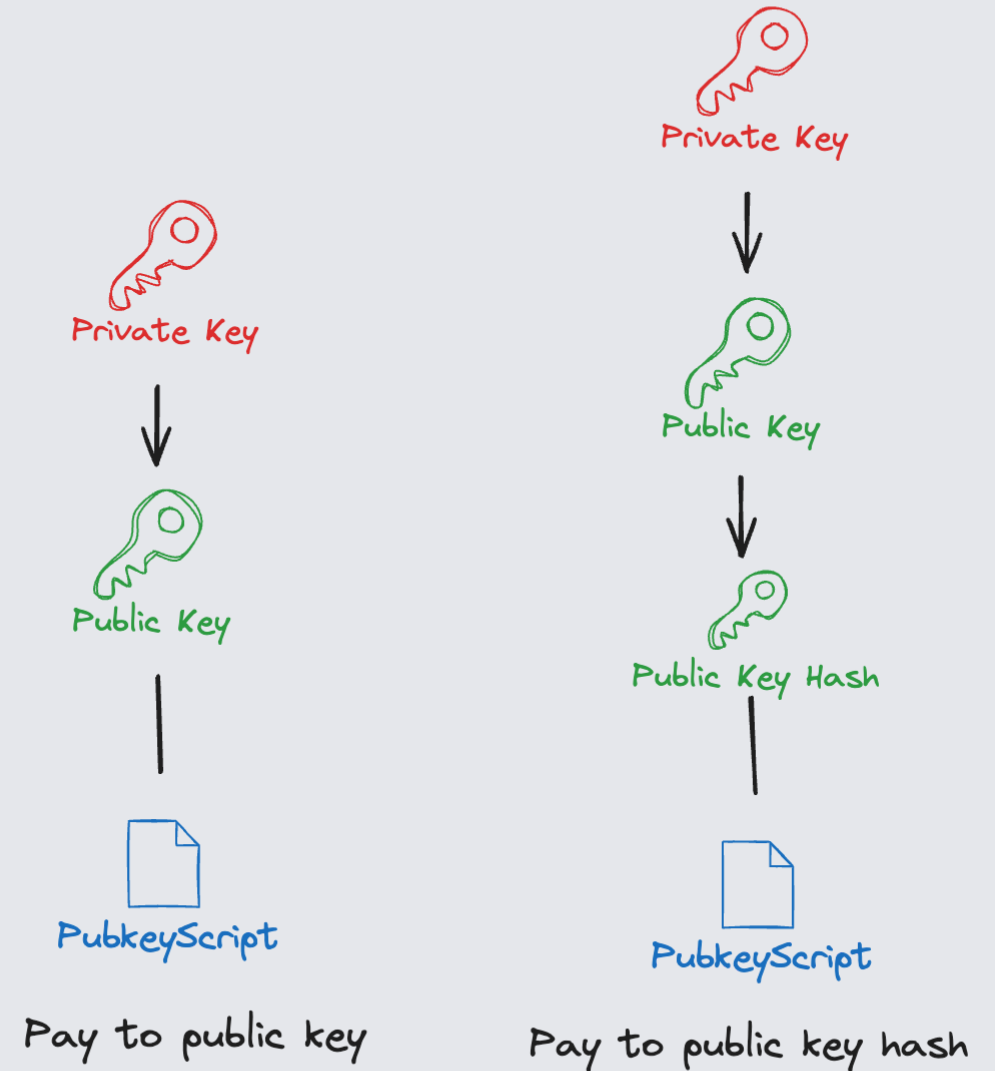
- [https://en.bitcoinwiki.org/wiki/Pay-to-Pubkey\\_Hash](https://en.bitcoinwiki.org/wiki/Pay-to-Pubkey_Hash)

Pubkey script:

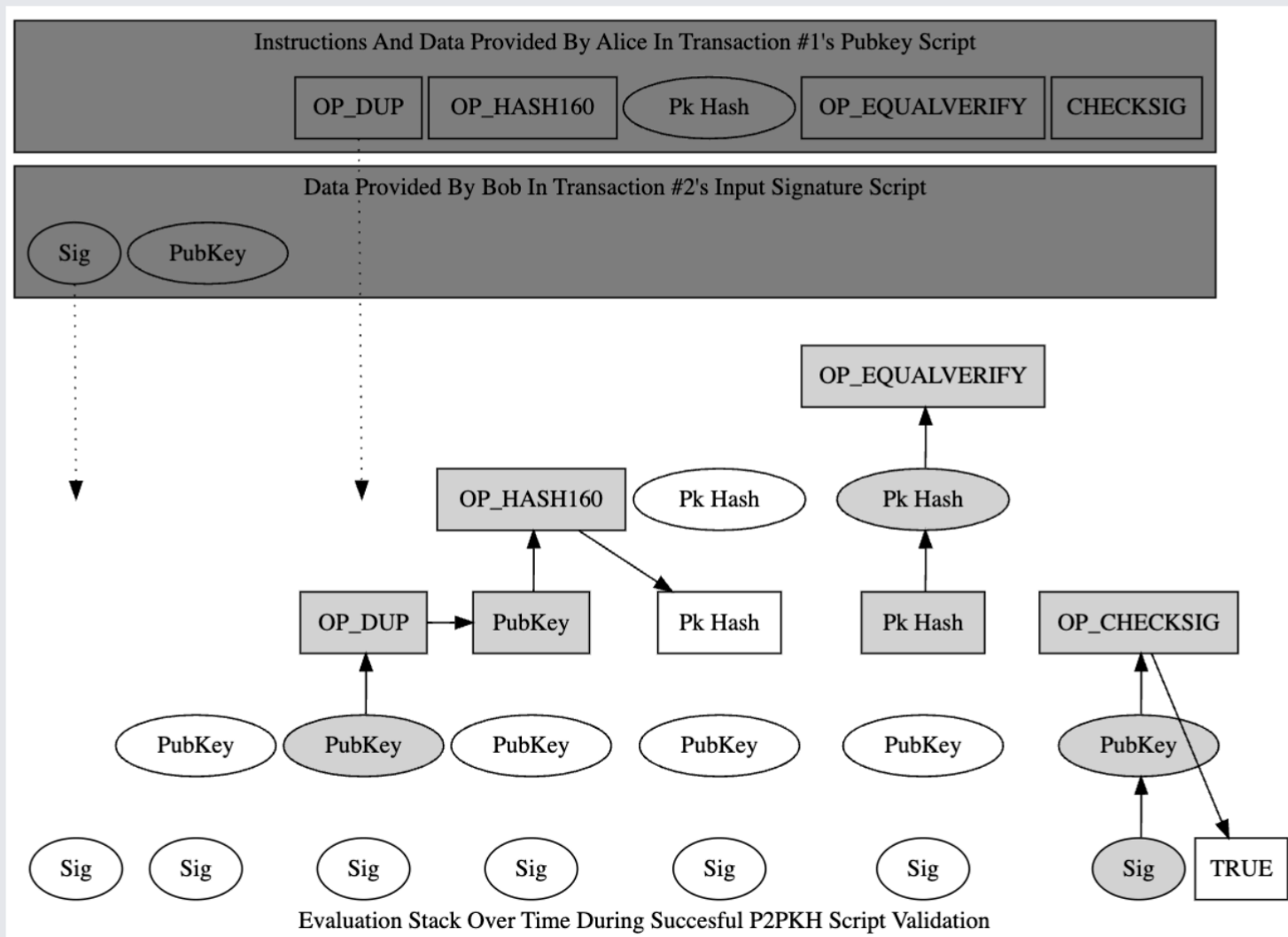
```
OP_DUP OP_HASH160 <PubkeyHash> OP_EQUALVERIFY OP_CHECKSIG
```

Signature script:

```
<Sig> <PubKey>
```



# P2PKH explained



# Pay-to-Script Hash (P2SH)

Payer can specify a redeem script.

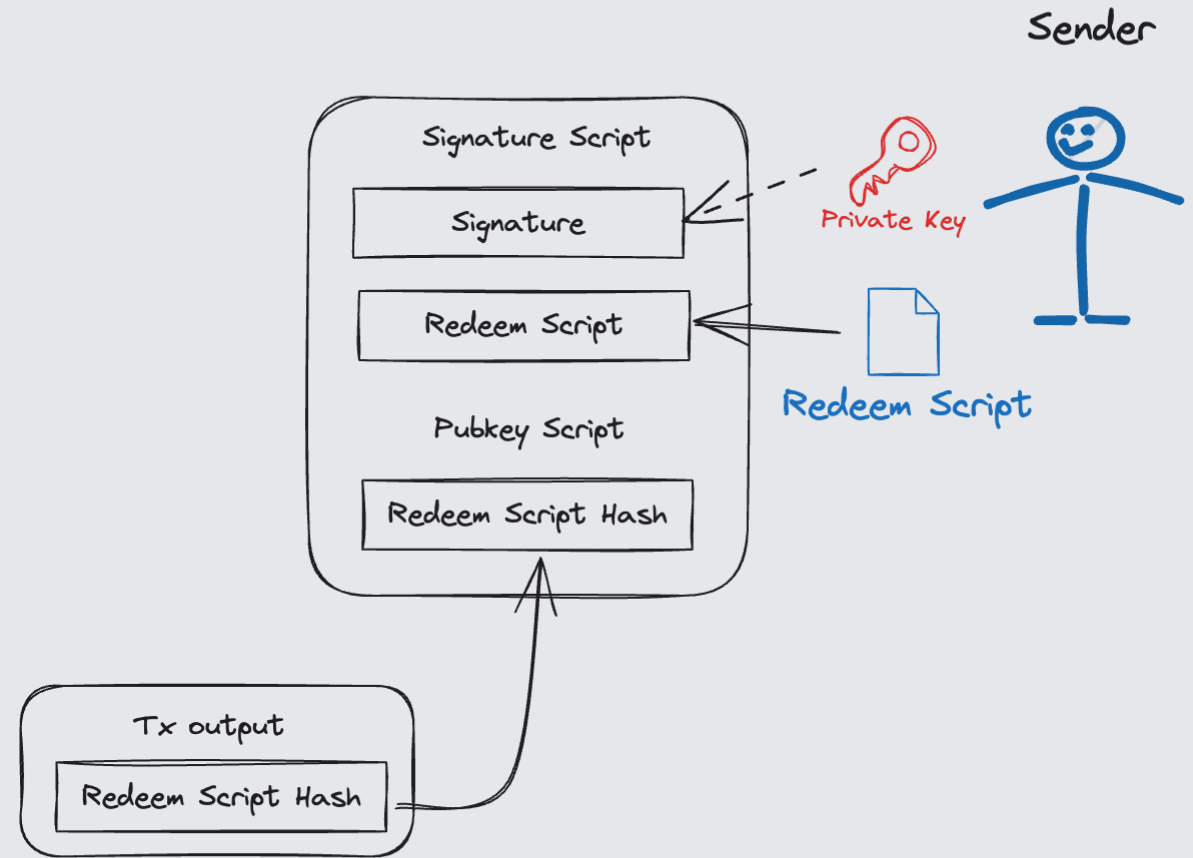
- [https://en.bitcoinwiki.org/wiki/Pay-to-Script\\_Hash](https://en.bitcoinwiki.org/wiki/Pay-to-Script_Hash)

Pubkey script:

```
OP_HASH160 <H(Redeem Script)> OP_EQUAL
```

Signature script:

```
<Sigs> <Redeem Script>
```



**Quiz time!**